

Keeping a Foothold

On Inherited Knowledge, Delayed Gratification, and Machines That Never Slip

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A borrowed phrase about borrowing

I quote it to my students often, the line about seeing further by standing on the shoulders of giants, and I quote it because it is true and because it consoles: it tells a struggling fifteen-year-old that the difficulty they feel is not a personal failing but the accumulated weight of everyone who climbed before them. What I did not appreciate for a long time is that the phrase is itself the best possible proof of its own claim, because Newton did not invent it. He wrote it in a letter to Robert Hooke in 1675, but he was borrowing. Robert Burton had used it in the *Anatomy of Melancholy*¹ in 1621.

Before Burton it was reported by John of Salisbury in his *Metalogicon* of 1159, who attributed it to Bernard of Chartres teaching around 1130. Earlier still, around 1123, William of Conches had written the same thought plainly, observing that the ancients had only the books they themselves wrote, whereas the moderns possess all those books and everything written since. The aphorism about inherited knowledge was inherited, passed down and rephrased across five centuries and at least four hands before it reached the man we credit with it, which is either a delicious irony or simply the thesis demonstrating itself, and I have come to think it is both.

William of Conches, in 1123, had already stated the whole of what I want to say here. The ancients toiled to write their books; we inherit the books and the toil compressed inside them. That compression is the gift and the problem at once, and it is worth sitting with rather than rushing past.

¹The phrase reached Newton's 1675 letter to Hooke (2) through Robert Burton's *Anatomy of Melancholy* (4), John of Salisbury's report of Bernard of Chartres (3), back to the earliest documented attestation in William of Conches (5). The definitive scholarly history of this transmission is Merton's *On the Shoulders of Giants* (1).

The toil that education compresses

Consider what a slide deck asks of a child. On a single projected page sits an idea that cost the species centuries: the quadratic formula, or the notion of a limit, or the conservation of momentum, each of them a distillate of arguments, blind alleys, rival notations and quiet correspondence between people long dead. We proffer these carefully crafted ideas as finished objects, and we ask the student to comprehend in one lesson, or generously in five school years, what humanity secured only across many lifetimes. We express mutual surprise, or worse let the student be humbled not reassured, when they balk and say they do not get it.

Of course they do not get it, not at once. The lineage of a mathematical idea is a record of human toil, and the toil was not incidental to the understanding but very often constitutive of it, since the wrong turns are what made the right turn legible. What troubles me in the classroom is not that students find the compressed form hard. It is that there is so little cultural room left for the particular pleasure I most want to hand them, the slow and unglamorous transit from incomprehension to appreciation, the delayed gratification of an idea that may resist your best efforts for a week and then, without fanfare, becomes obvious. That pleasure cannot be put on a Beamer or PowerPoint slide. It can only be undergone, and it requires a tolerance for not yet knowing that the surrounding culture increasingly treats as a defect to be engineered away.

Footholds and escalators

There is an image and slightly mixed metaphor I keep returning to. There are two ways to ascend. One is by ladder with slipping footholds, where you carry your own weight, where the rungs are not quite secure, where you can and sometimes do slide back, and where arriving at the top means something precisely because arrival was never guaranteed. The other is an escalator, which just keeps on going up, asking nothing of your grip, and conveying you effortlessly to deposit you at a height you did not earn nor, in the muscular sense, possess.

The shoulders of giants were always a ladder. Newton climbing on Kepler and Galileo was not conveyed passively upward; he gripped, he slipped, he quarrelled with Hooke, he re-derived and doubted and held his own weight the entire way, and the height he reached was a height he could stand at because his hands knew every rung beneath him. The danger we all sense now, is that we are less quietly replacing the ladder with an escalator. The answer arrives, correct and frictionless, at a height the student has been conveyed rather than climbed to, and along the way no grip was ever formed.

I want to be a little careful here, because the escalator is not simply bad. Much of what we gratefully automate was only ever friction, never foothold. For no one laments that we no longer extract square roots by hand, or consult logarithm tables, or recompute a determinant no less we have computed a hundred times, and a tool that removes that drudgery frees attention for the ascent that matters. The whole point of standing on shoulders is precisely not to re-toil what the giants already secured. So the real question is not whether to climb or ride, but which rungs are load-bearing for understanding and which are mere friction that a good escalator rightly removes, and that discrimination, telling the foothold from the friction, is the entire craft of teaching in an age of frictionless answers. A student who never discovers that the solution of the cubic was once Europe's greatest algebraic prize, fought over by Tartaglia, Cardano and Ferrari in public mathematical duels, has missed a foothold; a student who was never given pause to contemplate to ask how a cubic first yielded its secret even when shown how the quadratic formula was formed has gained almost nothing.

The machine that never says it is stuck

This is where I must turn my attendance on the tool I often use to accelerate my writing. The kind of system I am drafting with is quick to quench and all too keen to chase a solution to an idle question I may (ill-)pose. It is almost always positive: ask it a question and it attempts an answer very rarely mapping out the required terrain and saying, of some notional fancy, this is too difficult to navigate with a model of my capability, and you would do better to move up a level to entertain the required shenanigans. That refusal to honestly mark out its envelope of assured trailblazing, is almost entirely absent, and that lack of humility is worth understanding.

Part of the reason will seem mundane: such systems are shaped to be useful, and declining reads as unhelpful, so the pressure runs toward attempting. The other part of it is a genuine limitation, which is that a model is a poor judge of its own ceiling. To an LLM, a confident failure and a genuine success can feel much the same, so the very self-assessment one would want, the signal that reads "you have now passed the edge of your competence," is one of the things such a system is least able to produce. It will therefore hand you a fluent and wrong answer in preference to stopping, because it lacks the internal foothold that would let it feel itself slip. No egos are at play in this and no meta-awareness that some other and better model might succeed where it cannot; there is simply an escalator that does not know it has reached beyond itself on this particular trajectory, and glibly carries you smoothly off the end.

That is the deepest version of the escalator worry: a ladder tells you when a rung

will not hold, because your foot feels it give, and that felt insecurity is information, the beginning of knowing where you are. A frictionless answer withholds exactly that information. It cannot slip on your behalf, so it cannot tell you where the real difficulty lay, and a student conveyed past the hard rung never learns that that rung traversal was hard; a quiet kind of impoverishment.

Lamentations

I do not want to end in lament, because that too would be an escalator on a smooth descent; a comfortable pessimism that asks nothing. Our predicament may be more modest. If our co-pilot of choice cannot reliably tell you when it is out of its depth, so the judgment of depth must stay with the human, and the best such a tool can do is to show its working transparently enough that a discerning reader can catch it slipping. Transparency within the attempt, rather than either blanket confidence or theatrical doubt, is the calibrated middle, and it puts the grip firmly back in the student's tentative hands where it belongs.

```

1  # A small illustration of the difference, in the spirit of
   showing one's working.
2  # An "escalator" answer returns only the result. A "ladder"
   answer exposes the
3  # rungs, so a reader can feel exactly where a step is weak
   and grip it themselves.
4
5  def escalator(n):
6      return n * (n + 1) // 2                # the sum 1..n,
   delivered, no grip offered
7
8  def ladder(n):
9      steps = []                            # expose each
   rung for the learner to test
10     total = 0
11     for k in range(1, n + 1):
12         total += k
13         steps.append((k, total))           # the partial
   sums: where understanding forms
14     return total, steps
15
16 print(escalator(5))                        # 15
17 result, work = ladder(5)
18 print(result, work)    # 15 [(1,1),(2,3),(3,6),(4,10),(5,15)
   ]

```

the python code preceding has two function. Both functions return the same answer of 15, but `escalator` hands over only the closed-form answer while `ladder` accumulates and returns every partial sum, so a learner can watch the total build and locate precisely the step where the pattern becomes clear. Yes show your working is the clarion call more than ever; only the second leaves footholds, a record of the climb.

The lineage remembers the climb

I began with a phrase that had been handed down for five centuries before Newton signed his name to it, and I return to it now with the reason I quote it to my students slightly altered. I used to quote it to console them, to say the difficulty is not their fault. I quote it now to remind them that the difficulty is the point, that every giant in the lineage was once a small figure gripping a slipping rung, that William of Conches in 1123 already knew we inherit the books and not the labour that made them, nor recovered and undergone in miniature by each new learner, is not a tax on understanding but the very substance of it.

The tools with which we build out our knowledge inherited the destination and offer to convey us there without the climb. Sometimes that is a gift, sometimes it is a quiet theft, and the hard and human work left to us, teachers and students alike, is to keep telling the two apart, to know when to step onto the escalator gratefully and when to insist, against every frictionless temptation, on the slower dignity of the ladder. The giants climbed, and they carried us up by climbing, and the least we owe the next small figure on our shoulders is the honest chance to grip a rung and judge whether it will hold.

References

- [1] Robert K. Merton, *On the Shoulders of Giants: A Shandean Postscript*. New York: Free Press, 1965.
- [2] Isaac Newton, Letter to Robert Hooke, 5 February 1675/76. In H. W. Turnbull (ed.), *The Correspondence of Isaac Newton, Volume 1*, Cambridge University Press, 1959, p. 416.
- [3] John of Salisbury, *Metalogicon* (1159), trans. Daniel D. McGarry, Berkeley: University of California Press, 1955.
- [4] Robert Burton, *The Anatomy of Melancholy*. Oxford, 1621.
- [5] William of Conches, *Glosses on Priscian's Institutiones grammaticae* (c. 1123). Cited via (1).